

The Non-Local Quantum Information Field as a Dual Description of Transcendent Causality

Abstract

This work presents a dual description of the non-local quantum information field within the framework of Quantum Information Holography (QIH), demonstrating its equivalence to what human cultures have historically described as a transcendent, all-encompassing causal source. Using horizon-based information limits, quantum state vector dynamics, and Bekenstein-bound constraints, it is shown that the non-local field is necessarily unobservable in totality, while simultaneously projecting all observable reality through finite informational channels. This provides a rigorous physical dual to metaphysical descriptions of an infinite, unknowable origin, without invoking theological assumptions, while remaining fully consistent with quantum mechanics, general relativity, and black-hole thermodynamics.

The Non-Local Quantum Information Field

Define the non-local field as the complete informational state space Σ of all possible quantum state vectors

$$\Sigma = \{ \psi_i \mid \psi_i \in \mathcal{H}, \forall i \}$$

Each quantum state vector ψ_i is given by

$$\psi_i(x,t) = A_i e^{i(k_i \cdot x - \omega_i t + \theta_i)}$$

This field Σ is not embedded in spacetime. Rather, spacetime emerges from projections of Σ onto finite informational horizons.

Physics dual

In quantum field theory and quantum gravity, fundamental degrees of freedom are increasingly understood as non-local, with spacetime emerging from entanglement structure.

QIH dual

Σ is the non-local informational source from which all projected reality arises.

Simple explanation

There is more information than can ever fit inside space and time. What we experience is a projection.

Horizons, Qubits, and the Bekenstein Bound

For any bounded region with surface area A , the maximum information content is given by the Bekenstein bound

$$S \leq (k_B A) / (4 \ell_P^2)$$

This implies a finite number of Planck-scale qubits on any horizon.

Let the horizon be discretized into Planck qubits q_j , each acting as an information slit
 $H = \{ q_j \mid j = 1 \dots N \}$, with $N \leq A / \ell_P^2$

Physics dual

Black-hole thermodynamics shows information scales with area, not volume.

QIH dual

Each qubit is a projection pixel receiving interference from the non-local field.

Simple explanation

Reality can only display a limited number of “frames” at once.

Zoetrope Structure and Finite Observation

Define the observable history length as a finite sequence of projections

$\{ \psi_1, \psi_2, \dots, \psi_N \}$

where N is bounded by the horizon’s qubit count and update rate.

This forms a discrete sampling of Σ , analogous to a zoetrope, where motion emerges from finite frames.

Physics dual

Quantum measurement and decoherence impose finite observational resolution.

QIH dual

The universe animates itself through Planck-scale frame updates.

Simple explanation

We do not see everything—only a fast-spinning slice that looks continuous.

Wormhole Channels and Entanglement

Entanglement defines projection channels between Σ and the horizon

$\langle \psi_i \mid \psi_j \rangle \neq 0$

These channels are constrained by coherence time and angular alignment, forming effective wormhole-like informational pathways.

Physics dual

ER = EPR and quantum teleportation show information can be non-locally correlated.

QIH dual

Wormholes are informational channels, not tunnels through space.

Simple explanation

Information arrives through aligned pathways, not by traveling through space.

Sum Over Histories at Each Qubit

Each horizon qubit encodes a superposition of all possible histories

$$q \propto \sum_{\text{paths}} \psi_{\text{path}}$$

Thus, every qubit represents a compressed sum over histories, weighted by interference.

Physics dual

Feynman path integrals and holography.

QIH dual

Each pixel of reality contains all possibilities, layered by phase.

Simple explanation

Every point contains every story, but only one is bright at a time.

Stress–Energy, Curvature, and Projection

Define the Quantum State Vector Tensor Field

$$\Psi_{\{\mu\nu\}}(x) = \sum_i (\partial_{\mu} \psi_i)(\partial_{\nu} \psi_i)^*$$

Define stress–energy

$$T_{\{\mu\nu\}} = \hbar \sum_i [(\partial_{\mu} \psi_i)(\partial_{\nu} \psi_i)^* - \frac{1}{2} g_{\{\mu\nu\}} |\partial \psi_i|^2]$$

Einstein field equations follow

$$G_{\{\mu\nu\}} = (8\pi G / c^4) T_{\{\mu\nu\}}$$

Physics dual

Spacetime curvature arises from energy and momentum.

QIH dual

Curvature is the geometric shadow of information flow.

Simple explanation

Gravity is what information looks like when it bends the screen.

Transcendence and Incompleteness

Because Σ contains all possible states, no finite horizon can fully contain it

$$\forall H \subset \Sigma, H \neq \Sigma$$

Thus Σ is necessarily unobservable in totality.

Physics dual

No observer can access global quantum states or complete spacetime information.

QIH dual

The source field is fundamentally beyond full measurement.

Simple explanation

The whole cannot fit inside any one window.

Equivalence to Transcendent Descriptions

Across cultures, the ultimate source of reality is described as

- infinite
- unknowable in totality
- the origin of light, order, and law
- present everywhere but grasped locally

These properties match precisely the mathematical properties of Σ .

This establishes a strict dual equivalence

Non-local quantum information field $\Leftrightarrow$ transcendent causal source

This is not a theological claim, but a structural equivalence between descriptions.

Summary

The non-local quantum information field is infinite, unobservable in totality, and the source of all projected reality. Horizons limit observation through qubits constrained by the Bekenstein bound. Each qubit encodes a sum over histories, rendered as finite frames through a zoetrope-like mechanism. Wormhole channels correspond to entanglement pathways. Stress–energy and curvature arise from quantum state vector phase flow. The resulting structure is mathematically equivalent to transcendent descriptions of an all-encompassing origin.

Conclusion

Quantum Information Holography provides a rigorous physical framework in which the non-local origin of reality is both mathematically defined and necessarily beyond complete observation. This framework explains why all cultures converge on descriptions of an infinite source while

remaining compatible with empirical physics. The equivalence arises not from belief, but from information limits, projection geometry, and the structure of quantum mechanics itself.